

Visualising Learning Pathways in MOOCs

Karsten Øster Lundqvist
Victoria University of Wellington
Wellington, New Zealand
karsten.lundqvist@vuw.ac.nz

Steven Warburton
Victoria University of Wellington
Wellington, New Zealand
steven.warburton@vuw.ac.nz

Abstract—It is difficult to get an understanding of learner behaviour within MOOCs. Commonly used metrics are not well suited for this. This paper describes a visualisation tool based on modified Sankey diagrams to explore learner paths through edX MOOCs. The tool visualises learners' progression through the learning material and their interaction with discussion forums. These diagrams can be filtered based on user demographics and behaviours. Automated clustering of users based on behaviour metrics is available to provide suggestions of different learner behaviours. Statistical tests are also included to allow hypothesis exploration by researchers. The tool was developed using a devops development method to improve the usability of the tool for MOOC learning designers and researchers.

Index Terms—Data Visualization, Learner Behaviour, MOOC

I. INTRODUCTION

The level of active learner engagement has been the most significant contributor to the success of MOOCs [1]. However, the high level of learner numbers make it very difficult to follow individual progression through learning material, and therefore it is difficult for MOOC developers to understand actual learner behaviour within their courses and if the design of the MOOC leads to the expected learner behaviours. In structured xMOOCs it can be particularly difficult to get an overview of the effectiveness of the MOOC, and whether certain parts of a MOOC could be redesigned to improve it for the learners [2]. This paper describes a tool that can be used to visualise learner behaviour in MOOCs and provide insights for a better learning design.

II. VISUALISING LEARNER BEHAVIOUR IN MOOCs

Early research into learner behaviours and the perception of MOOCs from an individual learner perspective has primarily studied learners directly (e.g. through interviews or surveys), or focused on simple metrics (e.g. the number of videos viewed or number of tests taken) as proxies to then form an understanding of learner progress [3]–[6]. With the increase of the number of MOOCs over the years, a variety of metrics (e.g. number of sign ups, retention, or content views), comparing MOOCs have been used to measure “success” or learner satisfaction of courses [7]–[12]. These metrics mirror common metrics already used in higher education and are readily so perceived as easily understood and they are easy to calculate. However, they do not provide much information about actual learner behaviour within the courses and are therefore difficult to use for quality assurance and evaluation

purposes of individual courses. The value of these metrics as a quality measure of can also be questioned [13], [14]. Some research has gone beyond these metrics, often creating dashboards based on logs summarising user behaviours and interaction with the learning content [15]–[17].

In this paper we report on the use of Sankey diagrams as a novel visual tool to uncover learner behaviour. Sankey diagrams are used to visualise the flow of “traffic” from a number of states to other states [18]. Previously they have been used for instance to provide overviews of video consumption within MOOCs [19], [20]. In this case, a specialized type of Sankey state transition diagram was used to illustrate the number of users who viewed each part of a set of videos. Students who had watched all of the videos in succession could be identified and correlated with ‘qualified’ and ‘non-qualified’ status [21]. Google Analytics can provide standard Sankey diagrams to visualise transitions within websites, and these have been used to visualise behaviour within MOOCs [22]–[24]. However, these are session based and therefore one single user will be visualised multiple times when using a MOOC across several sessions [25]. This is a major problem as MOOCs typically run over several weeks and most users will use many sessions. Similar visualisations have been used, for instance using trajectory networks and graphs [26], to create overviews of learner paths or show individual progress, however these lack visualisation of the volume of student movement through the content.

III. BACKGROUND

A prototype tool was developed at Victoria University of Wellington (VUW) to visualise individual learner movements, including the volume, within their edX hosted MOOCs. Fig. 1 shows the original visualisation. Each vertical bar is a page of content in an edX course. The upper line of vertical bars represents the first visit to a page, while the lower line represents every consecutive visit. The grey horizontal bars (or flow bars) that connect from one page to the next are the expected movements, i.e. learners are following the expected linear route through the learning material where they move from one page to the next. The blue flow bars show forward jumps, i.e. the participant are skipping pages, and the green flow bars are backward jumps to earlier content, i.e. revisiting a page. The red flow bars indicate the last user activity i.e. where the participant stopped following the course. For example in fig. 1 it is clear that many participants skipped the

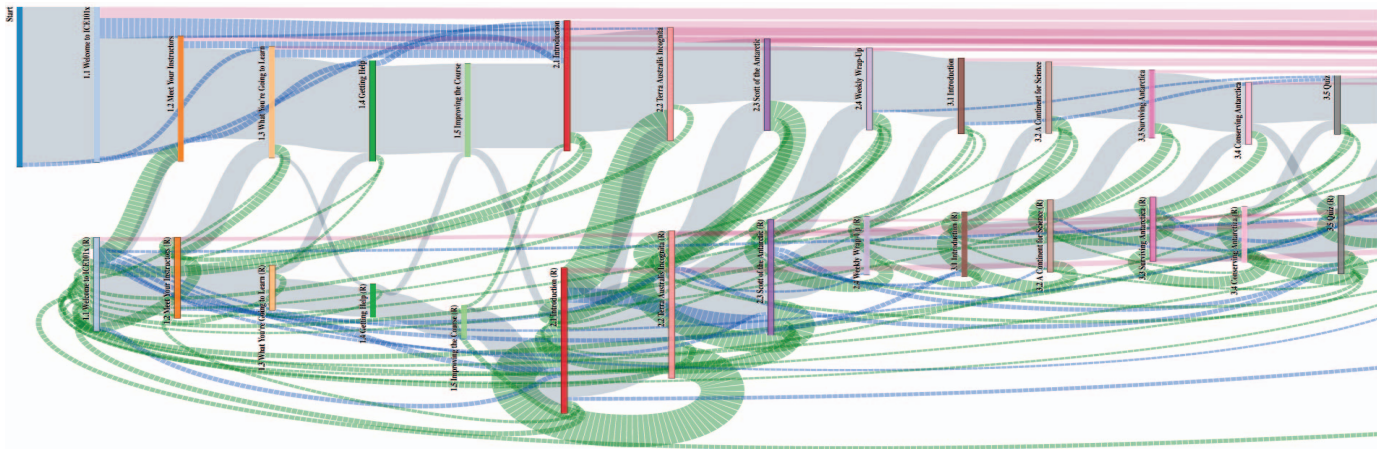


Fig. 1. The original visualisation.

introduction pages and went straight to the main content, while many jumped back from various places, to the first page of main content. The prototype tool described here was developed in an agile fashion with iterative testing by MOOC developers and stakeholder groups within VUW. Overall, it was found to be a useful tool to better understand learner behaviour and was useful in surfacing unexpected transitions through MOOCs under study. Additionally, the stakeholder groups also indicated several possible extensions, such as filtering, statistics and clustering, that could be useful improvements to the effectiveness of the visualisation tool [2].

IV. METHOD

The prototype tool was further developed successively by two Master level students, for six and then three months full-time. They both followed a “devops” methodology [27] where they developed functionalities which were immediately reviewed by learning designers from the MOOC team to provide instant feedback and help speed of development toward a useful product. The first developer spent approximately 50% of his development time with the learning designer, and the second spent 100% of her development time with the designer. Through this process the usefulness of the visualisation tool was validated in an iterative fashion, incorporating feedback from the MOOC learner designers and additionally the MOOC steering group at VUW, which comprised MOOC researchers, administrators and practitioners.

The visualisations presented in this paper use data from the various MOOCs run at VUW on the edX platform. This is not a study of a particular group of MOOC, therefore details about these are omitted (together with detail within some of the figures). The emphasis in this paper is on the visualisation and the functionalities available for learning designers within the tool.

V. FUNCTIONALITIES

This section will describe the functionalities of the tool, how it has evolved, and insights that have been captured in the process.

A. Filtering

The most requested extension was the ability to filter the view based on groupings of participants. The administrators and researchers in particular were interested in being able to view how different cohorts interacted with the courses, in order to spot differences that would stimulate further study.

The tool was therefore extended to provide filtering of any demographic information that could be retrieved from the edX logs. This included gender, age, educational level, date viewed (both before, after, and during specific periods), enrolment dates of students (e.g. to study differences in behaviour between participants who enrol early and late), payment status (paid for a certificate or not) and certificate status (received or not). Furthermore, a custom filter functionality was created where a separate csv file with user ids could be provided and used for filtering.

After receiving positive feedback on the extended filter functionality it was suggested that the developer should add filters based on data captured in the pre-course surveys. These were added using the data exported from the Qualtrics survey system.

A filtering functionality was also embedded into the Sankey view whereby the user can double-click flow bars of special interest to immediately filter the view based on the users captured in that step. Therefore it became possible to investigate what users, who do something specific in one part of the course, are doing in other places.

B. Statistics

The researchers within the MOOC steering group were interested in the integration of statistical testing for the visualisations, to establish if certain behaviours in a specific filtered cohort were statistically significant to everybody else.

The statistics experts at VUW reviewed the visualisation tool and indicated the absence of an appropriate statistical tool to test the significance of difference of multiple paths. However, they determined that it would be appropriate to embed chi-squared tests for individual flow paths within the



Fig. 2. Filtering and statistic

view, as long as the test within the visualisations are not used to test hypotheses. The statistics can be used as hypothesis gathering, i.e. finding interesting areas to further explore for evidence elsewhere. The reasoning for this, is that as the datasets originated from the MOOCs it will always be possible to find significant differences in the datasets. However it is impossible to say if these are reproducible or something happening only in the visualised dataset.

Fig. 2 shows an example of filtering showing only movements by all known females. The p-level has been set at 0.05. The opaque flow paths are the steps with a p-value below 0.05. In this example there is a statistically significant difference that suggests that female MOOC users are more likely to revisit pages several times and do so several times. This cannot be used to prove that this is a behaviour by females on MOOCs, through a study could be made to investigate why this happens. Is it universal? Is it something specific with the content of pages that makes it more interesting or important for female participants? This, a researcher could use as observations to start new exploration of the data or as hypothesis for further studies. The values of the chi squared tests are shown as a tooltip when the user hovers over the flow paths (fig. 3).

C. User interface improvements

Over time, working with the MOOC development team has helped to improve the readability of the visualisation. Fig. 2 highlights how the vertical page bars are now better aligned when comparing to those in fig. 1. In addition, the content bars can now be separated further by using a slider. The view has also been extended to show a fine granularity within the visualisation. The interface that is presented to the user when starting the visualisation tool itself has also been enhanced via several feedback cycles.

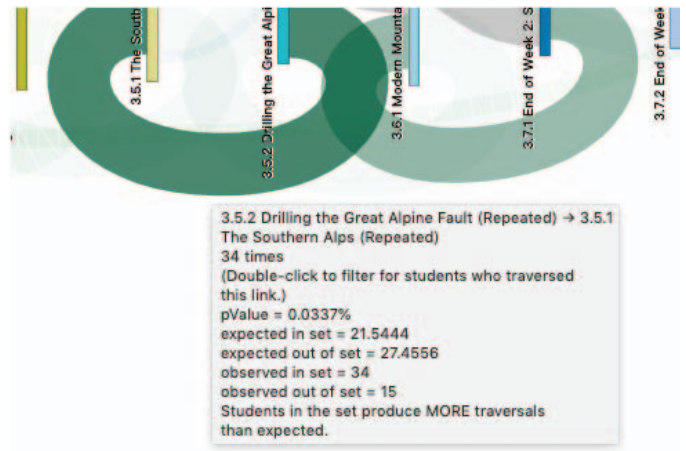


Fig. 3. Chi squared test

D. Videos

Video usage statistics were also integrated into the view as secondary information in the tooltip. However, the learning designer team did not find any value in this approach as the standard analytics tools available through the edX platform already show this information. Several alternative views involving direct visualisation within the Sankey diagram were attempted. In the diagram the main user-interaction elements are content pages, but as each video is part of a page there was no user-friendly way to show this intuitively, without cluttering the current view, was found. This was furthermore complicated by the need to distinguish fully versus partially viewed videos. The decision was therefore made to omit this information from the view.

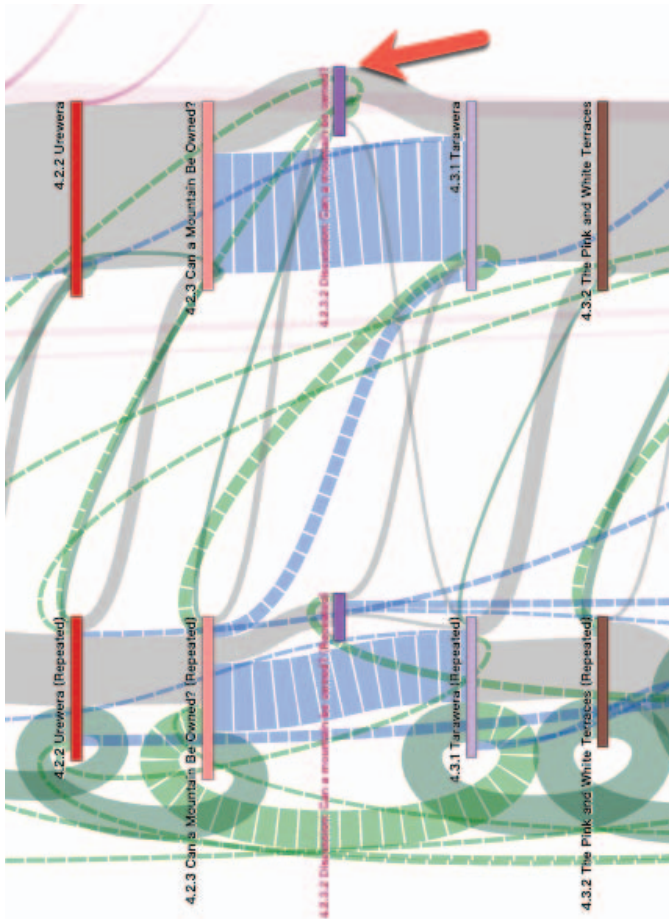


Fig. 4. Forum represented in the sankey diagram

E. Forums

The success of many MOOCs relies on the active engagement of the learners. Through their dynamic engagement with the course, learners self-organise into learning communities where they are able to share skills, objectives, knowledge, and interests, through comments within discussion forums [1]. Based on this observation there was a strong support from the academics on the MOOC steering group and learning designer team to incorporate forum activity into the Sankey diagram. This was achieved (Fig. 4) by adding additional vertical bars for each of the forums available on the individual courses. These bars are shown slightly raised when compared to the regular learning material bars to make them visually distinct (as show by the arrow). Hovering over a forum bar a tooltip is displayed with the information relating to the number of views, posts and replies on the particular forum (Fig. 5). Feedback from both the academics and the learning design team was critical to enhancing the usability of this part of the tool.

F. Clustering

The data set that is generated based on learner progression and interaction is big and it is difficult to create an overview of

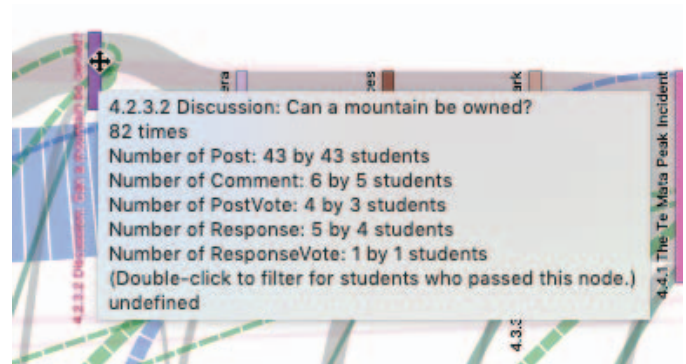


Fig. 5. Forum represented in the sankey diagram

various styles of behaviour. One suggestion was to incorporate clustering functionality to facilitate researchers who wants to analyse learner behaviours within MOOCs. Clustering algorithms are a set of algorithms that explore data with no prior knowledge and identify similar data objects within a particular data set [28]. This was integrated into the visualiser using the Weka data mining library [29].

The user selects this clustering option before starting the visualiser (see fig. 6). There are a selection of clustering algorithms to choose from within the tool and the number of clusters that are required must be selected. Finally, the user needs to select the data types of most interest when running the clustering algorithm. Currently these cluster cover:

- expected moves / total number of moves
Fraction of moves that are from one step to the next out of all the moves made, i.e. the expected moves.
- (forward moves - expected moves) / total number of moves
Fraction of moves that are not to the expected move and backward movements. Therefore the fractions of forward jumps in the learning content.
- backward moves / total number of moves
Fraction of movements to previous learning content.
- first time visits / total number of visits
Fraction of all visits to learning content that are to previously unseen material.
- Revisits / total number of visits
Fraction of all visits to previously seen material
- Forum visits / total number of visits
Fraction of all visits to forums
- posts / statements
Fraction of all statement made on forums that are new posts. Statements include posts, comments and replies to comments.
- Statements made / total number of views
An indication of forum activity on forums compared to other activity.

When the user is starting the visualisation there are in this example three clusters available together with the standard visualisation of all learners. It is also possible to get an overview of all the clusters at once. Fig. 7 shows an overview

Fig. 6. Selecting clustering

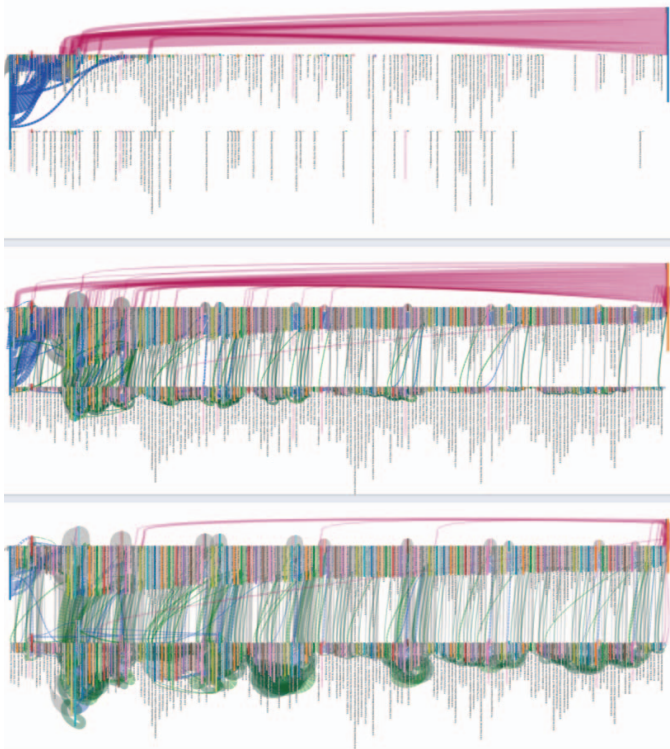


Fig. 7. High-level overview of clusters

of three clusters from a single MOOC. It is worth noting that even though the clusters are made based on metrics unrelated to retention the clusters have clearly identified two behaviour types that are more likely to leave and one that follow the MOOC to the end.

G. Using the tool

The tool has already been used within the learning design team and by MOOC academics, for example to identify issues which the designers should address in future runs of courses. For instance, it has been observed that many participants skip introductory information and jump straight to the main content. Therefore, placing important information at the beginning should be avoided and re-positioned where possible within the teaching content that follows. Some content pages appear to disrupt the expected course flow, with learners stopping and exiting the course. This observation provides a prompt for designers to examine the content and flow at specific points

in a course and modify where needed to improve the design. The team have also noticed that tests can be disruptive. Some learners simply skip them, and others use them as strategic learning guides to jump backwards and forwards between tests and content to answer the questions. Therefore, constructive alignment of tests and learning objectives is important, though some developers are more relaxed as they view learning in MOOCs as more informal.

VI. FUTURE DEVELOPMENT

Although the visualisation tool in its current format has been developed to be used with logs from the edX platform, the underlying data structures have been developed to be platform agnostic i.e. there is a log reader that translate the edX logs into a data structure which is then used by the visualiser. This means that, in theory, it should be relatively easy to use other MOOC platform logs, with a similar granularity of information (content level, learner interaction information) and translate those into the same data structures. The team is interested in doing this and it is the plan to open source the tool to enable others to achieve this as desired.

VII. CONCLUSION

This paper has described the development of a tool that uses Sankey diagrams to successfully visualise learner behaviours in MOOCs. The tool was developed and validated through an iterative user centred design methodology. The DevOps approach, placing the developer and end user together to work closely through each iteration, was critical to the enhancement of both the embedded suite of functionalities and the user experience. It has been designed and tested with two core user groups: learning designers and educational researchers. For learning designers the tool has proved valuable in linking learning analytics to course design. The tool helps complete a loop whereby meaningful data is returned to those professionals engaged in creating large online courses and can be used to both validate learning designs and to redesign course elements where necessary [30], [31]. For researchers the tool has provided insights in novel patterns of behaviour that have previously been difficult to uncover. By providing a set of functionalities to test statistical significance and to explore clustering, it has the potential to offer researchers a platform from which to differentiate and further investigate learner participation in large courses. The next phase of this project will be to extend testing of the tool out to the wider educational and research community and bring new insights to further develop the visualisation and analysis functionalities.

REFERENCES

- [1] A. McAuley, B. Stewart, G. Siemens, and D. Cormier, "The MOOC model for digital practice," http://www.elearn.space.org/Articles/MOOC_Final.pdf Accessed 6 June 2017, 2010.
- [2] K. Lundqvist, M. Godinez, and S. Warburton, "Visualizing learner behaviour in moocs using sankey diagrams," in *Open Oceans: Learning without borders. Proceedings ASCILITE*, M. Campbell, J. Willems, C. Adachi, D. Blake, I. Doherty, S. Krishnan, S. Macfarlane, L. Ngo, M. O'Donnell, S. Palmer, L. Riddell, H. Sure, and J. Tai, Eds. Geelong: ASCILITE, 2018, pp. 194–203.

- [3] R. Kop and H. Fournier, "New dimensions to self-directed learning in an open networked learning environment," *International Journal of Self-Directed Learning*, vol. 7, no. 2, pp. 1–18, 2010.
- [4] R. Kop, H. Fournier, and J. S. F. Mak, "A pedagogy of abundance or a pedagogy to support human beings? participant support on massive open online courses," *The International Review Of Research In Open And Distributed Learning*, vol. 12, no. 7, pp. 74–93, 2011.
- [5] D. Levy, "Lessons learned from participating in a connectivist massive online open course (MOOC)," in *Proceedings of the Chais conference on instructional technologies research 2011: Learning in the technological era*, 2011, pp. 31–36.
- [6] B. Stewart, "Social media literacies and perceptions of value in open online courses," http://portfolio.cribchronicles.com/wp-content/uploads/2012/11/612701_Social_Media_Literacies_MOOCs.pdf Accessed 7 July 2017, 2010.
- [7] P. Adamopoulos, "What makes a great MOOC? an interdisciplinary analysis of student retention in online courses," in *Proceedings of the 34th International Conference on Information Systems*, Milan, 2013, pp. 1–21.
- [8] K. Jordan, "MOOC completion rates: The data," <http://www.katyjordan.com/MOOCproject.html> Accessed 7 June 2017, 2013.
- [9] H. Khalil and M. Ebner, "MOOCs completion rates and possible methods to improve retention-a literature review," in *World Conference on Educational Multimedia, Hypermedia and Telecommunications*, 2014, pp. 1305–1313.
- [10] K. Jordan, "Massive open online course completion rates revisited: Assessment, length and attrition," *The International Review of Research in Open and Distributed Learning*, vol. 16, no. 3, pp. 341–358, 2015.
- [11] D. F. Onah, J. Sinclair, and R. Boyatt, "Dropout rates of massive open online courses: behavioural patterns," *EDULEARN14 Proceedings*, pp. 5825–5834, 2014.
- [12] K. Stephens-Martinez, M. A. Hearst, and A. Fox, "Monitoring moocs: which information sources do instructors value?" in *Proceedings of the first ACM conference on Learning@ scale conference*. ACM, 2014, pp. 79–88.
- [13] T. R. Liyanagunawardena, P. Parslow, and S. Williams, "Dropout: Mooc participants perspective," in *EMOOCs 2014, the Second MOOC European Stakeholders Summit*, Lausanne, Switzerland, 2014.
- [14] T. R. Liyanagunawardena, K. Ø. Lundqvist, P. Parslow, and S. A. Williams, "MOOCs and retention: does it really matter?" in *Can MOOCs save Europe's unemployed youth? EC-TEL*, September 2014.
- [15] R. Cobos, S. Gil, A. Lareo, and F. A. Vargas, "Open-dlas: an open dashboard for learning analytics," in *Proceedings of the Third (2016) ACM Conference on Learning@ Scale*. ACM, 2016, pp. 265–268.
- [16] D. Davis, I. Jivet, R. F. Kizilcec, G. Chen, C. Hauff, and G.-J. Houben, "Follow the successful crowd: raising mooc completion rates through social comparison at scale," in *Proceedings of the seventh international learning analytics & knowledge conference*. ACM, 2017, pp. 454–463.
- [17] S. Grover, R. Pea, and S. Cooper, "Promoting active learning & leveraging dashboards for curriculum assessment in an openedx introductory cs course for middle school," in *Proceedings of the First ACM Conference on Learning @ Scale Conference*, ser. L@S '14. New York, NY, USA: ACM, 2014, pp. 205–206. [Online]. Available: <http://doi.acm.org/10.1145/2556325.2567883>
- [18] M. Schmidt, "The sankey diagram in energy and material flow management," *Journal of industrial ecology*, vol. 12, no. 1, pp. 82–94, 2008.
- [19] C. Shi, S. Fu, Q. Chen, and H. Qu, "Vismooc: Visualizing video clickstream data from massive open online courses," in *Visual Analytics Science and Technology (VAST), 2014 IEEE Conference on*. IEEE, 2014, pp. 277–278.
- [20] H. Qu and Q. Chen, "Visual analytics for mooc data," *IEEE computer graphics and applications*, vol. 35, no. 6, pp. 69–75, 2015.
- [21] C. Coffrin, L. Corrin, P. de Barba, and G. Kennedy, "Visualizing patterns of student engagement and performance in moocs," in *Proceedings of the Fourth International Conference on Learning Analytics And Knowledge*, ser. LAK '14. New York, NY, USA: ACM, 2014, pp. 83–92. [Online]. Available: <http://doi.acm.org/10.1145/2567574.2567586>
- [22] S. R. Emmons, R. P. Light, and K. Börner, "Mooc visual analytics: Empowering students, teachers, researchers, and platform developers of massively open online courses," *Journal of the Association for Information Science and Technology*, vol. 68, no. 10, pp. 2350–2363, 2017.
- [23] T. Beaven, T. Codreanu, and A. Creuzé, "Motivation in a language mooc: issues for course designers," *Language MOOCs: Providing Learning, Transcending Boundaries*. Berlin: De Gruyter Open, pp. 48–66, 2014.
- [24] J. Kay, P. Reimann, E. Diebold, and B. Kummerfeld, "Moocs: So many learners, so much potential..." *IEEE Intelligent Systems*, vol. 28, no. 3, pp. 70–77, 2013.
- [25] G. Analytics, "Using the flow visualization reports," 2017, <https://support.google.com/analytics/answer/2519989>.
- [26] M. Ginda, M. C. Richey, M. Cousino, and K. Börner, "Visualizing learner engagement, performance, and trajectories to evaluate and optimize online course design," *PloS one*, vol. 14, no. 5, p. e0215964, 2019.
- [27] C. Ebert, G. Gallardo, J. Hernantes, and N. Serrano, "Devops," *IEEE Software*, vol. 33, no. 3, pp. 94–100, May 2016.
- [28] R. Xu and D. C. Wunsch, "Survey of clustering algorithms," 2005.
- [29] M. Hall, E. Frank, G. Holmes, B. Pfahringer, P. Reutemann, and I. H. Witten, "The weka data mining software: an update," *ACM SIGKDD explorations newsletter*, vol. 11, no. 1, pp. 10–18, 2009.
- [30] L. Corrin, N. Law, U. L. Ringtved, and S. Milligan, "Evaluating systems and tools that link learning analytics and learning design," in *The 8th International Learning Analytics & Knowledge Conference*. Society for Learning Analytics Research, 2018.
- [31] M. Schmitz, E. Van Limbeek, W. Greller, P. Sloep, and H. Drachsler, "Opportunities and challenges in using learning analytics in learning design," in *European Conference on Technology Enhanced Learning*. Springer, 2017, pp. 209–223.